FUNDAMENTALS OF LIGHT SOURCES AND LASERS

Mark Csele

WILEY-INTERSCIENCE

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To my parents
for fostering and encouraging
my interest in science

CONTENTS

PREFACE

The field of photonics is enormously broad, covering everything from light sources to geometric and wave optics to fiber optics. Laser and light source technology is a subset of photonics whose importance is often underestimated. This book focuses on these technologies with a good degree of depth, without attempting to be overly broad and all-inclusive of various photonics concepts. For example, fiber optics is largely omitted in this book except when relevant, such as when fiber amplifiers are examined. Readers should find this book a refreshing mix of theory and practical examples, with enough mathematical detail to explain concepts and enable prediction of the behavior of devices (e.g., laser gain and loss) without the use of overwhelmingly complex calculus. Where possible, a graphical approach has been taken to explain concepts such as modelocking (in Chapter 7) which would otherwise require many pages of calculus to develop.

This book, targeted primarily to the scientist or engineer using the technology, offers the reader theory coupled with practical, real-world examples based on real laser systems. We begin with a look at the basics of light emission, including blackbody radiation and atomic emission, followed by an outline of quantum mechanics. For some readers this will be a basic review; however, the availability of background material alleviates the necessity to refer back constantly to a second (or third) book on the subject. Throughout the book, practical, solved examples founded on real-world laser systems allow direct application of concepts covered. Case studies in later chapters allow the reader to further apply concepts in the text to real-world laser systems.

The book is also ideal for students in an undergraduate course on lasers and light sources. Indeed, the original design was for a textbook for an applied degree course (actually, two courses) in laser engineering. Unlike many existing texts which cover this material in a single chapter, this book has depth, allowing the reader to delve into the intricacies. Chapter problems assist the reader by challenging him or her to make the jump between theory and reality. The book should serve well as a text for a single course in laser technology or two courses where a laboratory component is present. Introductory chapters on blackbody radiation, atomic emission, and quantum mechanics allow the book to be used without the requirement of a second or third book to cover these topics, which are often omitted in similar texts. It is assumed that students will already have a grasp of geometric and wave optics (including the concepts of interference and diffraction), as well as basic first-year physics, including kinematics.

Chapter 1 begins with a look at the most basic light source of all, blackbody radiation, and includes a look at standard applications such as incandescent lighting as well as newer applications such as far-IR viewers capable of "seeing" a human body hidden in a trunk! Chapter 2 is a look at atomic emission, in which we examine the nature and origins of emission of light from electrically excited gases as well as mechanisms such as fluorescence, with applications ranging from common fluorescent lamps to vacuum-fluorescent displays and colored neon tubes. The chapter concludes with a look at semiconductor light sources (LEDs). This chapter also includes atomic emission theory (such as the origins of line spectra) as well as practical details of spectroscopy, including operating principles of a spectroscope and examples of its use in identifying unknown gas samples, which serve to reinforce with practical applications the usefulness of the entire theory of atomic structure.

Investigation of both blackbody radiation as well as atomic emission light leads us on a path to quantum mechanics (in Chapter 3), vital to understanding the mechanisms responsible for light emission at an atomic level and later for understanding the origins of transitions responsible for laser emission in the ultraviolet, visible, and infrared regions of the spectrum. Although few books include this topic, it is vital to understanding emission spectra as well as basic laser processes, and so is included in the book for some readers as a review, for others as a new topic.

In Chapter 4 we begin with a fundamental look at lasers and lasing action. Aside from the basic processes, such as stimulated emission and rate equations governing lasing action, we also outline key laser mechanisms, such as pumping, the requirement for feedback (examined in detail in Chapter 6), and gain and loss in a real laser. Real-world examples are embedded within the chapter, such as noise in a fiber amplifier, which demonstrates the rate equations in action as well as details of an experiment in which the gain of a gas laser is measured by insertion of a variable loss into the optical cavity. In Chapter 5 we examine lasing transitions in detail, including selective pumping mechanisms and laser energy-level systems (three- and four-level lasers). Examples of transitions and energies in real laser systems are given along with theoretical examples, allowing the reader to compare how well theoretical models fit real laser systems.

In addition to expanding the concepts of gain and loss introduced in Chapter 4, in Chapter 6 we examine the laser resonator as an interferometer. The mathematical requirements for stability of a resonator and longitudinal and transverse modes in a real resonator are detailed. Wavelength selection mechanisms, including gratings, prisms, and etalons, are outlined in this chapter, with examples of applications in practical lasers, such as single-frequency tuning of a line in an argon laser.

Chapter 7 provides the reader with an introduction to techniques used to produce fast pulses such as Q-switching and modelocking. In the case of modelocking a graphical approach is used to illustrate how a pulse is formed from many simultaneous longitudinal modes. In Chapter 8 we cover nonlinear optics as they apply to lasers. Harmonic generation and optical parametric oscillators are examined.

The last six chapters of the book provide case studies allowing the reader to see the practical application of laser theory. The lasers chosen represent the vast majority of commercially available lasers, allowing the reader to relate the theory learned to practical lasers that he or she encounters in the laboratory or the manufacturing environment. In these chapters various lasers are outlined with respect to the lasing process involved (including quantum mechanics, energy levels, and transitions), details of the laser itself (lasing medium, cooling requirements), power sources for the laser, applications, and a survey of commercially available lasers of that type. Visible gas lasers, including helium–neon and ion lasers, are covered in Chapter 9. UV gas lasers such as nitrogen and excimer lasers are covered in Chapter 10, including details of the unique constraints on electrical pumping for these types of lasers. In Chapter 11 we examine infrared gas lasers focusing primarily on carbon dioxide and similar lasers using rotational and vibrational transitions.

Chapter 12 details common solid-state lasers, including YAG and ruby. Pump sources, including flashlamps, CW arc lamps, and semiconductor lasers, are examined as well as techniques such as nonlinear harmonic generation (often used with these lasers). Chapter 13 details the basics of semiconductor lasers, and Chapter 14 covers dye lasers which feature wide continuously tunable wavelength ranges and are often used in modelocking schemes to generate extremely short pulses of laser radiation. In each of these case study chapters, photographs and details have been included, allowing the reader to see the structure of each laser. Chapter 10, for example, includes numerous photographs of the various structures in a real excimer laser, including details of the electrodes and preionizers, the cooling system, and heat exchanger, as well as electrical components such as the energy storage capacitor and thyratron trigger. In each case a clear explanation is given guiding the reader to understanding the function of each critical component.

Adjunct material to this text, including in-depth discussions of spectroscopy (with color photos of example spectra), analysis and photographic details of real laser systems (such as real ion, carbon dioxide, and solid-state lasers), additional problems, and instructional materials (such as downloadable MPEG videos) covering practical details such as cavity resonator alignment may be found on the author's host web site at http://technology.niagarac.on.ca/lasers.

I must thank a number of people who have made this book possible. First, Dr. Marc Nantel of PRO (Photonics Research Ontario) for instigating this in the first place—it was his suggestion to write this book and he has been extremely supportive throughout, reviewing manuscripts and providing invaluable input. I would also like to thank colleague and fellow author Professor Roy Blake, who has helped coach me through the entire process; director of the CIT department at Niagara College, Leo Tiberi, who has been supportive throughout and has provided an environment in which creative thought flourishes; Dr. Johann Beda of PRO for reviewing the manuscript and for suggestions on material for the book; my counterpart at Algonquin College, Dr. Bob Weeks, for suggestions and illuminating discussions on excimer laser technology; and assistant editor Roseann Zappia of John Wiley & Sons. In addition to providing reviews that provided excellent feedback on the material, Roseann did a remarkable job of simplifying the entire publishing process and has gone out of her way to make this endeavor as painless as possible. I'd also like to thank my wife (who has become a widow to my laptop computer for the past year and a half) for her patience and support all along.

Finally, I would like to acknowledge not only Niagara College but especially my photonics students in various laser engineering courses for their review of my

manuscripts (many under the guise of course notes) and providing invaluable input from a student's point of view. Many students provided critical insight into proofs and problems in this book which would only be possible from an undergraduate. In some cases they highlighted difficult concepts that required clarification.

I welcome the opportunity to hear from readers, especially those with suggestions for improving the book. Please feel free to e-mail me at mcsele@ieee.org. Since I get a large volume of e-mail (and spam), please refer to the book in the subject line.

MARK CSELE

CHAPTER 1

Light and Blackbody Emission

As a reader of this book, you are no doubt familiar with the basic properties of light, such as reflection and interference. This book deals with the *production* of light in its many forms: everything from incandescent lamps to lasers. In this chapter we examine the fundamental nature of light itself as well as one of the most basic sources of light: the blackbody radiator. Blackbody radiation, sometimes called *thermal light* since the ultimate power source for such light is heat, is still a useful concept and governs the workings of many practical light sources, such as the incandescent electric lamp. In later chapters we shall see that these concepts also form a base on which we shall develop many thermodynamic relationships which govern the operation of other light sources, such as lasers.

1.1 EMISSION OF THERMAL LIGHT

We have all undoubtedly encountered *thermal light* in the form of emission of light from a red-hot object such as an element on an electric stove. Other examples of such light are in the common incandescent electric lamp, in which electrical current flowing through a thin filament of tungsten metal heats it until it glows white-hot. The energy is supplied by an electrical current in what is called resistance heating, but it could just as well have been supplied by, say, a gas flame. In fact, the original incandescent lamp was developed in 1825 for use in surveying Ireland and was later used in lighthouses. The lamp worked by spraying a mixture of oxygen and alcohol (which burns incredibly hot) at a small piece of lime and igniting it. The lime was placed at the hottest part of the flame and heated until it glowed white-hot, emitting an immense quantity of light. It was the brightest form of artificial illumination at the time and was claimed to be 83 times as bright as conventional gas lights of the time. Improvements to the lamp were made by using a parabolic reflector behind the piece of lime concentrating the light. The lamp allowed the surveying of two mountain peaks over 66 miles apart and was later improved by using hydrogen and oxygen as fuel but eventually was superseded by the more convenient electric arc lamp. This light source did, however, find its way into theaters, where it was used as a

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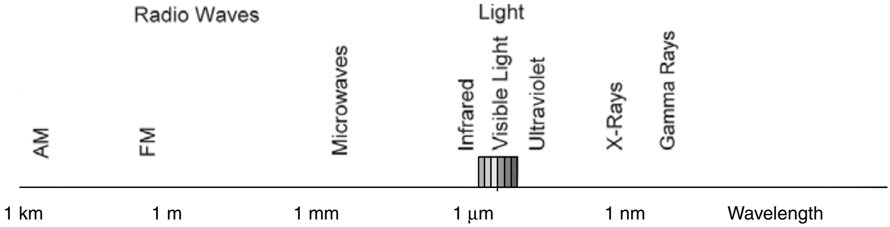

Figure 1.2.1 Electromagnetic spectrum.

spotlight which replaced the particularly dangerous open gas flames used at the time for illumination, and hence the term *limelight* was born.[1]

1.2 ELECTROMAGNETIC SPECTRUM

Anyone with a knowledge of basic physics knows that light can be viewed as a particle or as a wave, as we shall examine in this chapter. Regardless of the fact that it exhibits particlelike behavior, it surely does exhibit wavelike behavior, and light and all other forms of electromagnetic behavior are classified based on their wavelength. In the case of visible light, which is simply electromagnetic radiation visible to the human eye, the wavelength determines the color: Red light has a wavelength of about 650 nm and blue light has a wavelength of about 500 nm. Electromagnetic radiation includes all forms of radio waves, microwaves, infrared radiation, visible light, ultraviolet radiation, x-rays, and gamma rays. Figure 1.2.1 outlines the entire spectrum, including corresponding wavelengths.

1.3 BLACKBODY RADIATION AND THE STEFAN–BOLTZMANN LAW

Imagine a substance that absorbs all incident light, of all frequencies, shining on it. Such an object would reflect no light whatsoever and would appear to be completely black—hence the term *blackbody*. If the blackbody is now heated to the point where it glows (called *incandescence*), emissions from the object should, in theory, be as perfect as its absorption—one would logically expect it to emit light at all frequencies since it absorbs at all frequencies. In the 1850s a physicist named Kirchhoff, a pioneer in the use of spectroscopy as a tool for chemical analysis, observed that real substances absorb better at some frequencies than others. When heated, those substances emitted more light at those frequencies. The paradox spawned research into radiation in general and specifically, how radiation emitted from an object varied with temperature.

[1]An excellent historical description of the events surrounding the development of the limelight as well as the arc lamp is presented in the *Connections* series hosted by James Burke (BBC, 1978).

It was observed that the amount of radiation emitted from an object varied with the temperature of the object. The mathematical relationship for this dependence on temperature was established in 1879 by the physicist Josef Stefan, who showed that the total energy radiated by an object increased as the fourth power of the temperature of the object. All objects at a temperature above absolute zero (0 K) radiate energy, and when the temperature of an object is doubled, the total amount of energy radiated from the object will be 16 times as great!

In 1884, Ludwig Boltzmann completed the mathematical picture of a blackbody radiator, and the Stefan–Boltzmann law was developed, which allows calculation of the total energy integrated over a blackbody spectrum.

$$W = \sigma T^4 \tag{1.3.1}$$

where σ is the Stefan–Boltzmann constant ($=5.67 \times 10^{-8}\ \mathrm{W/m^2 \cdot K^4}$) and T is the temperature in kelvin. This law applies, strictly speaking, to ideal blackbodies. For a nonideal blackbody radiator, a third term, called the *emissivity* of an object, is added, so the law becomes

$$W = e\sigma T^4 \tag{1.3.2}$$

where the emissivity of the object (e) is a measure of how well it radiates energy. An ideal blackbody has a value of 1; real objects have a value between 0 and 1.

Example 1.3.1 Use of the Stefan–Boltzmann Formula The Stefan–Boltzmann formula gives an answer in watts per square meter. This can be rearranged to give the power output of an object as

$$P = A\sigma T^4$$

where A is the surface area of the object. Now consider two objects: The first is a large (1-$\mathrm{m^2}$) object at a relatively cool temperature of 300 K. The total power radiated from this (ideal) object is 459 W. Now consider a much smaller ($10\ \mathrm{cm^2} = 1 \times 10^{-4}\ \mathrm{m^2}$) object at 3000 K. The total power radiated from this object is also 459 W. Although 10,000 times smaller, the object is also much hotter, so radiates a great deal of power.

Perhaps the most startling revelation from all this is that the 1-$\mathrm{m^2}$ object at room temperature emits 459 W at all. This may seem like an enormous amount of energy, especially when compared to a 500-W floodlight; however, essentially all of this output is in the far-infrared region of the spectrum and is manifested as heat. The human body, similarly, emits a fair quantity of heat (hence the reason for large air conditioners in office buildings that house large quantities of bodies in a relatively confined space).

1.4 WEIN'S LAW

Aside from the fact that radiated energy increases with temperature, it was also observed that the wavelengths of radiation emitted from a heated object also change as an object is heated. Objects at relatively low temperatures such as 200°C (473 K) do not glow but do indeed emit something—namely, infrared radiation—felt by human beings as heat. As an object is heated to about 1000 K, the object is seen to glow a dull-red color. We call this object "red hot." As the temperature increases, the red color emitted becomes brighter, eventually becoming orange and then yellow. Finally, the temperature is high enough (like that of the sun at 6000 K) that the light emitted is essentially white (white hot).

In an attempt to model this behavior, physicists needed an ideal blackbody with which to experiment; however, in reality, none exists. In most cases a nonideal material radiated energy in a pattern of wavelengths, depending on the chemical nature of the material. The physicist Wilhelm Wein, in 1895, thought of a way to produce, essentially, a perfect blackbody in a cavity radiator. His thought was to produce a heated object with a tiny hole in it which opens to an enclosed cavity as in Figure 1.4.1. Light incident on the hole would enter the cavity and be absorbed by the irregular walls inside the cavity—essentially a perfect absorber. Light that did reflect from the inner walls of the cavity would eventually be absorbed by other surfaces that it hit after reflection, as evident in the figure. If the entire cavity is now placed in a furnace and heated to a certain temperature, the radiation emitted from the tiny hole is blackbody radiation.

Wein's studies of cavity radiation showed that regardless of the actual temperature of an object, the pattern of the emission spectrum always looked the same, with the amount of light emitted from a blackbody increasing as wavelengths became shorter, then peaking at a certain wavelength and decreased rapidly at yet still shorter wavelengths in a manner similar to that shown in Figure 1.4.2 (in this example for an object at a temperature of 3000 K). In the figure, a peak emission is evident around 950 nm in the near-infrared region of the spectrum. Emission is also seen throughout the visible region of the spectrum (from 400 through

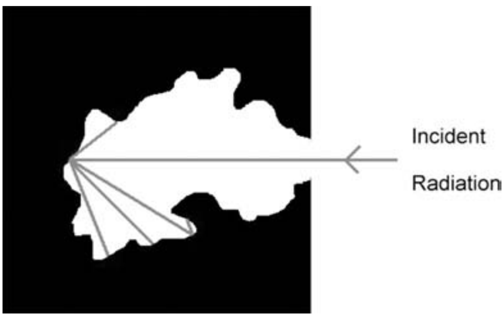

Figure 1.4.1 Cavity radiator absorbing incident radiation.

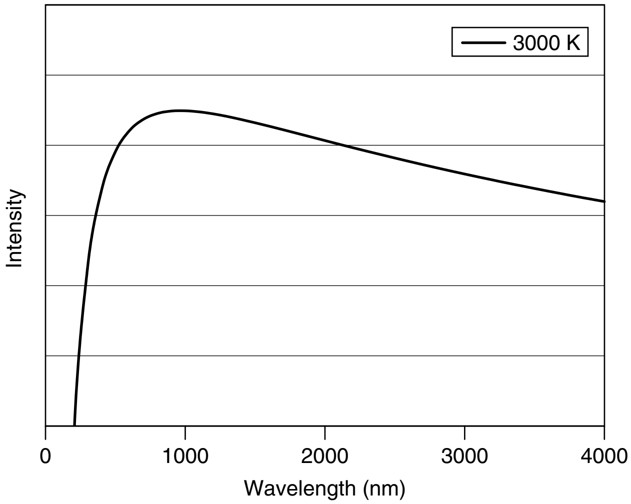

Figure 1.4.2 Typical blackbody output for an object at 3000 K.

700 nm), but more intensity is seen in the red than in the blue. It was observed that when the temperature was increased, the total amount of energy was increased according to the Stefan–Boltzmann law, and the wavelength of peak emission also became shorter. Wein's law predicts the wavelength of maximum emission as a function of object temperature:

$$\lambda_{\max} T = 2.897 \times 10^{-3}\ \mathrm{m} \cdot \mathrm{K} \tag{1.4.1}$$

where $\lambda_{\max}$ is the wavelength where the peak occurs and T is the temperature in kelvin. As an example, contrast the 3000 K object to the sun, which has essentially a blackbody temperature of 5270 K. The formula gives the peak at 550 nm in the yellow-green portion of the visible spectrum. This is indeed the most predominant wavelength in sunlight and is (presumably through evolutionary means) the wavelength of maximum sensitivity of the human eye.

Wein's law allows the prediction of temperature based on the wavelength of peak emission and may be used to estimate the temperature of objects such as hot molten steel as well as stars in outer space. A white star such as our sun has a temperature of around 5000 K, whereas a blue star is much hotter, with a temperature around 7000 K. Hotter stars are thought to consume themselves more quickly than cooler stars.

Both the Stefan–Boltzmann law and Wein's law may be verified experimentally using an incandescent lamp connected to a variable power source. As a practical thermal source an incandescent light does not exhibit perfect blackbody emission; however, it does follow the same general behavior as that of a blackbody source. Consider the actual spectral output of a 25-W incandescent light bulb measured

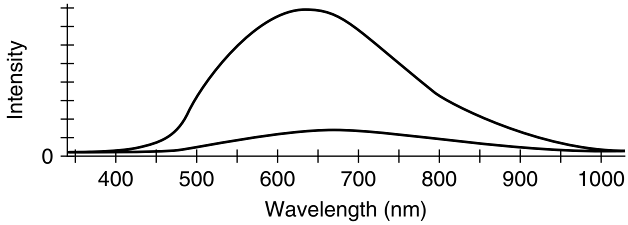

Figure 1.4.3 Output of an incandescent lamp.

both at full power and at 58% of full power (Figure 1.4.3). At full power the lamp exhibits maximum output around 625 nm, whereas at 58% of full power (the lower trace in the figure) the wavelength of peak emission shifts to around 660 nm, exhibiting the behavior predicted by Wein's law. The effect of filament temperature on intensity is also evident in the figure. Visually, light emitted from the lamp is observed to be more orange when the lamp is operated at the lower power (and hence, cooler filament) and quite white when operated at full power. We would expect white light to be consistent across the entire visible spectrum, having an intensity at 400 nm comparable to that at 600 nm, but this is clearly not the case here, and it may be surprising that this light which we call white is very rich in red and orange light (600 to 700 nm) and relatively weak in violet and blue light (400 to 500 nm). We shall revisit this idea later in the chapter.

1.5 CAVITY RADIATION AND CAVITY MODES

To derive a mathematical expression to describe the blackbody radiation curve (e.g., in Figure 1.4.2), many approaches were taken. As it is difficult to compute the behavior of a real blackbody, an easier avenue was found by Wein in the form of a cavity radiator, in which we assume that the object is a heated (isothermal) cavity with a hole in it from which light is emitted. Inside the enclosure the absorption of energy balances with emission. The cavity itself may be seen as a resonator in which standing waves are present. To simplify the problem further (to permit mathematical analysis), consider a cubical heated cavity of dimensions L (Figure 1.5.1).

A standing wave can be produced in any one of three dimensions, and standing waves of various wavelengths are possible as long as they fit inside the cavity (i.e., are an integral multiple of the dimensions of the cavity). The condition exists, then, that any electromagnetic wave (e.g., a light wave) inside the cavity must have a node at the walls of the cavity. As frequency increases, more nodes will fit inside the cavity, as evident in Figure 1.5.1. Rayleigh and Jeans showed mathematically that the number of modes per unit frequency per unit volume in such a cavity is

$$\frac{8\pi\nu^2}{c^3} \tag{1.5.1}$$

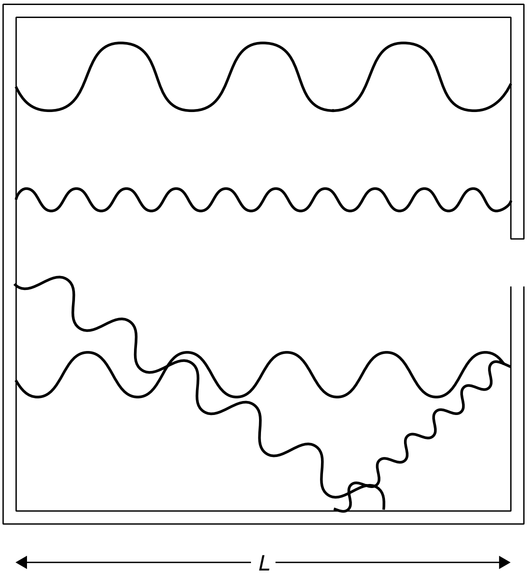

Figure 1.5.1 Modes in a heated cavity.

where ν is the frequency of the mode and c is the speed of light. Rayleigh and Jeans went on to postulate that the probability of occupying any given mode is equal for all modes (in other words, all wavelengths are radiated with equal probability), an assumption from classical wave theory, and that the average energy per mode was kT (from Boltzmann statistics). The last assumption was made according to the classical prequantum (i.e., before the Rutherford atomic model) view of radiation, in which each atom with an orbiting electron was considered to be an oscillator continually emitting radiation with an average energy of kT. The resulting law formulated by Rayleigh and Jeans describes the intensity of a blackbody radiator as a function of temperature as follows:

$$\text{intensity} = \frac{8\pi\nu^2}{c^3}kT \tag{1.5.2}$$

The law works well at low frequencies (i.e., long wavelengths); however, at higher frequencies it predicts an intense ultraviolet (UV) output, which simply was not observed. For any given object, 16 times as much energy observed as red light should be emitted as violet light, and this simply does not happen. This law predicts the equilibrium intensity to be proportional to $kT\nu^2$, a result that may be arrived at using classical electromagnetic theory and which states that an oscillator will radiate energy at a rate proportional to ν^2. This failure of the Rayleigh–Jeans law, dubbed the *UV catastrophe*, illustrates the problems with applying classical physics to the domain of light and why a new approach was needed.

A different approach to the problem was developed by Max Planck in 1899. His key assumption was that the energy of any oscillator at a frequency ν could exist only in discrete (quantized) units of $h\nu$, where h was a constant (called *Planck's constant*). The fundamental difference in this approach from the classical approach was that modes were quantized and required an energy of $h\nu$ to excite them (more on this in the next section). Upper modes, with higher energies, were hence less likely to be occupied than lower-energy modes.

Bose–Einstein statistics predict the average energy per mode to be the energy of the mode times the probability of that mode being occupied. Central to this idea was that the energy of the actual wave itself is quantized as $E = h\nu$. This is not a trivial result and is examined in further detail in Chapter 2, where experimental proof of this relation will be given. Multiplying this energy by the probability of a mode being occupied (the Bose–Einstein distribution function) gives the average energy per mode as

$$\frac{h\nu}{\exp(h\nu/kT) - 1} \tag{1.5.3}$$

where h is Planck's constant, ν the frequency of the wave, k is Boltzmann's constant, and T is the temperature. When this is multiplied by the number of modes per unit frequency per unit volume, the same number that Rayleigh and Jeans computed, the resulting formula allows calculation of the intensity radiated at any given wavelength for any given temperature:

$$\frac{8\pi h\nu^3}{kT}\left[\exp\left(\frac{h\nu}{kT}\right) - 1\right]^{-1} \tag{1.5.4}$$

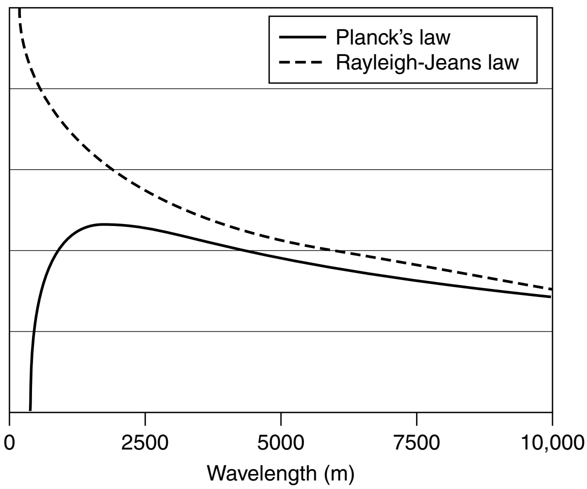

Figure 1.5.2 UV catastrophe and Planck's law.

Comparing Planck's radiation formula to the classical Rayleigh–Jeans law, we see that the two results agree well at low frequencies but deviate sharply at higher frequencies, with the Rayleigh–Jeans law predicting the UV catastrophe that never occurred (Figure 1.5.2). This new approach, which fit experimentally obtained data, heralded the birth of *quantum mechanics*, which we examine in greater detail in Chapters 2 and 3.

1.6 QUANTUM NATURE OF LIGHT

One of the earliest views of what light is was provided by Isaac Newton early in the eighteenth century. Based on the behavior of light in exhibiting reflection and refraction, he postulated that light was a stream of particles. Although this explanation works well for basic optical phenomena, it fails to explain interference. Later experiments, such as Young's dual-slit experiment, showed that light did indeed have a wavelength and that such behavior could only be explained by using wave mechanics, a concept Newton had argued against a century earlier. By the end of the nineteenth century, wave theory was well accepted but a few glitches remained: namely, the blackbody spectrum of light emitted from heated objects and issues such as the UV catastrophe, upon which classical physics failed. To explain this, Max Planck (the "father" of quantum mechanics) used particle theory once again. He postulated that the atoms in a blackbody acted as tiny harmonic oscillators, each of which had a fundamental quantized energy that obeyed the relationship

$$E = h\nu \tag{1.6.1}$$

where h is Planck's constant and ν is the frequency of the radiation emitted. This important equation may also be expressed in terms of wavelength as

$$E = \frac{hc}{\lambda} \tag{1.6.2}$$

where λ is the wavelength in meters. The ramifications of *quantization*—the fact that the energy of each atom was in integral multiples of this quantity—has far-reaching ramifications for the entire field of physics (and chemistry), and as we shall see in subsequent chapters, affects our entire view of the atom! Einstein also endorsed this concept of quantization and used it to explain the photoelectric effect, which definitely showed light to exhibit particle properties. These particles of light came to be known as photons, and according to the relationship above, were shown to have a discrete value of energy and frequency (and hence, wavelength). Light can be thought of as a wave that has particlelike qualities (or, if you prefer, a particle with wavelike qualities). It was evident from numerous experiments that both wave and particle properties are required to fully explain the behavior of light.

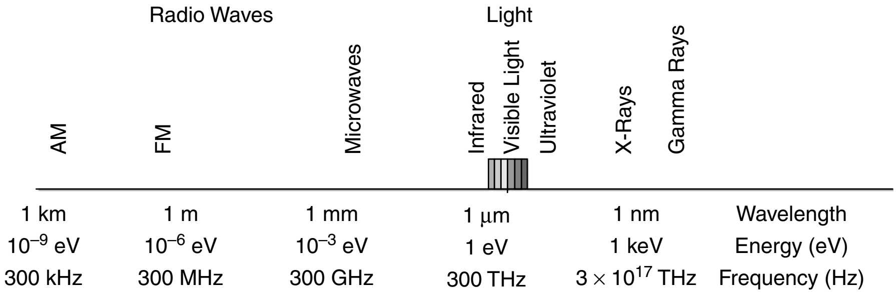

Figure 1.7.1 Electromagnetic spectrum with energies.

1.7 ELECTROMAGNETIC SPECTRUM REVISITED

Applying the photon model to the electromagnetic spectrum from Section 1.2, we may now find that a photon at a particular wavelength has a distinct energy. For example, photons of green light at 500 nm have an energy of

$$E = h\nu = \frac{hc}{\lambda}$$

or 3.98×10^{-19} J. More commonly, we express this energy in electron volts (eV), defined as the energy that an electron has accumulated after accelerating through a potential of 1 V. It is a convenient measure since the 500-nm photon can now be expressed as having an energy of 2.48 eV (with $1\ \text{eV} = 1.602 \times 10^{-19}$ J). Radio waves, microwaves, and infrared radiation have low energies, ranging up to about 1.7 eV. The visible spectrum consists of red through violet light, or 1.7 through 3.1 eV. UV, x-rays, and gamma rays have increasing photon energies to beyond 1 GeV. The spectrum is shown in Figure 1.7.1 with energies as well as wavelengths shown.

1.8 ABSORPTION AND EMISSION PROCESSES

Light is a product of quantum processes occurring when an electron in an atom is excited to a high-energy state and later loses that energy. Imagine an atom into which energy is injected (the method may be direct electrical excitation or simply thermal energy provided by raising the temperature of the atom). The electron acquires the energy and in doing so enters an excited state. From that excited state the electron can lose energy and fall to a lower-energy state, but energy must be conserved during this process, so the difference in energy between the initial high-energy state and the final low-energy state cannot be destroyed; it appears either as a photon of emitted light or as energy transferred to another state or atom. This is simply the principle of *conservation* of energy. The fact that atoms and molecules have such energy levels and transitions can occur between these

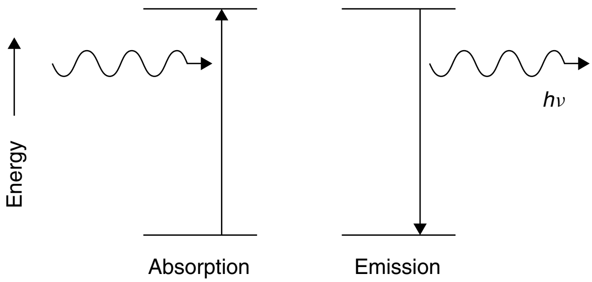

Figure 1.8.1 Absorption and emission of energy.

levels must be accepted for now: compelling experimental evidence is given in Chapter 2.[2]

An atom at a low-energy state can absorb energy and in so doing will be elevated to a higher-energy state. The energy absorbed can be in almost any form, including electrical, thermal, optical, chemical, or nuclear. The difference in energy between the original (lower) energy state and the final (upper) energy state will be exactly the energy that was absorbed by the atom. This process of *absorption* serves to excite atoms into high-energy states. Regardless of the excitation method, an atom in a high-energy state will certainly fall to a lower-energy state since nature always favors a lower-energy state (i.e., the law of entropy from thermodynamics). In jumping from a high- to a low-energy state, photons will be produced, with the photon energy being the difference in energy between the two atomic energy states. This is the process of *emission* (Figure 1.8.1). By knowing the energies of the two atomic states involved, we may predict the wavelength of the emitted photon using Planck's relationship according to

$$E_{\text{photon}} = E_{\text{initial}} - E_{\text{final}} = h\nu_{\text{photon}}$$

Examining an incandescent light bulb, we have a thin filament of tungsten metal glowing white-hot and emitting light. The energy is supplied in the form of an electrical current passing through the filament. In doing so, the filament is raised in temperature to 3000 K or more (the atmosphere inside a light bulb must be void of oxygen or the filament would quickly burn). This thermal energy excites tungsten atoms in the filament to high energy levels, but nature favors low-energy states, so the atoms will not stay in these high-energy states indefinitely. Soon, excited atoms will lose their energy by emitting light and falling to lower-energy states. The amount of energy lost by the atom appears as light. The low-energy atoms are now free to absorb thermal energy and to begin the process again. As long as energy is supplied to the system, tungsten atoms will continue this process of absorption of energy (in this case, thermal energy) and emission of light. Not all energy emitted is

[2]As we shall see in Chapter 2, gases such as hydrogen exhibit well-defined energy levels. Demonstrations such as the Franck–Hertz experiment prove that such energy levels exist at discrete, well-defined levels.

in the form of visible light. The largest portion of the electrical energy injected into the system appears as heat and infrared light.

Various forms of light use various means of excitation. In a neon sign, for example, an electrical discharge (usually at high voltages) pumps atoms to high-energy states. Gases usually have well-defined energy levels (as for the example of hydrogen in Figure 1.8.2), so their emissions often appear as *line spectra*: a series of discrete emissions (lines) of well-defined wavelengths. In the case of a fluorescent tube, a gas discharge emits radiation that excites phosphor on the wall of the tube. The phosphor then emits visible light from the tube. We examine these sources in detail, in Chapter 2, as they also provide insight into quantization. Other common forms of light include chemical glow sticks (the type you bend to break an inner tube to start them glowing). A reaction between two chemicals in the tube (previously separated but now mixed) excites atoms to high-energy states. In this chapter we examine primarily *thermal light* in which excitation is brought about solely by temperature.

To reiterate, all atoms and molecules have energy levels such as those depicted in Figure 1.8.2, which shows the levels for a hydrogen atom. Light is produced when an atom at a higher energy level loses energy and falls to a lower level in what is called a *radiative transition* (so called because radiation is emitted in the transition). Two such transitions are shown on the figure, one emitting red light and the other emitting blue light. Energy in such a transition must be conserved, so the difference in energy between the upper and lower levels is released as a photon of light. The larger the transition, the more energy the photon will have (e.g., a blue photon has more energy than a red photon). As well as emitting photons, there are also *nonradiative transitions* in which energy is lost without emission of radiation (e.g., as heat). These do not contribute to radiation emission, so are omitted from this discussion.

Evident in Figure 1.8.2 is a level labeled the *ground state* of the atom. This is the lowest energy state that an atom or molecule can have. All atoms and molecules

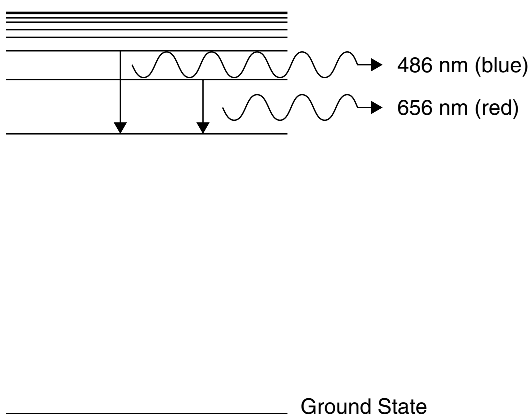

Figure 1.8.2 Simplified hydrogen energy levels.

assume this state at absolute zero (0 K), but frequently, the addition of energy (from electrical or thermal sources) causes the energy of the atom to rise to an upper level. The radiative transitions in Figure 1.8.2 are between two upper energy levels in the atom but could also be between an upper energy level and the ground state. In this case the transition would result in a higher-energy photon than those shown in the figure, and ultraviolet radiation would result.

Knowing where the energy levels are in a particular atom (and these are well known for essentially all atomic and molecular species), the problem now is to determine how many atoms are at a particular energy state. To do this, we use a thermodynamic concept called the *Boltzmann energy distribution*.

1.9 BOLTZMANN DISTRIBUTION AND THERMAL EQUILIBRIUM

Every system has thermal energy. In the case of a gas, consider gas atoms confined to a cylinder. Thermal energy manifests itself as atoms colliding with each other and bouncing off the cylinder walls. It is these collisions of gas atoms with the cylinder walls that are manifested as gas pressure. In a solid, the thermal energy manifests itself as vibrations of the atoms making up the solid; when the vibrations are too much for the interatomic forces keeping the atoms in place, the atoms are no longer tied to a particular spot and start flowing past one another; this is what happens when a solid melts.

Thermal energy also excites atoms (no matter what form they are in: gas, solid, or liquid), raising them to higher energy levels: The more thermal energy that is injected into a system (i.e., the higher the temperature), the more that higher energy levels will be populated. The resulting distribution of energy, describing the population of atoms at each energy level, is governed by Boltzmann's law, one of the fundamental laws of thermodynamics. Boltzmann's law predicts the population of atoms at a given energy level as follows:

$$N = N_0 \exp\left(-\frac{E}{kT}\right) \tag{1.9.1}$$

where N is the population of atoms at the given energy level, N_0 the population of atoms at ground state, E the energy above ground level in Joules, k is Boltzmann's constant (1.38×10^{-23} J/K), and T is the absolute temperature in Kelvin. This is to say that the law predicts an exponential drop in populations of atoms at higher energies. If we assume that a substance has an infinite number of energy levels (a continuum), the population at any given energy can be calculated using this law. The resulting distribution of energy for various energy levels is graphed in a generalized form in Figure 1.9.1.

We can rewrite Boltzmann's law as a ratio of N/N_0, allowing us to predict the distribution of atoms at any given energy level. Consider a generic substance at room temperature (293 K). In order to emit red light, we would need to excite atoms in the substance to 1.7 eV above ground (where an electron volt is equal to

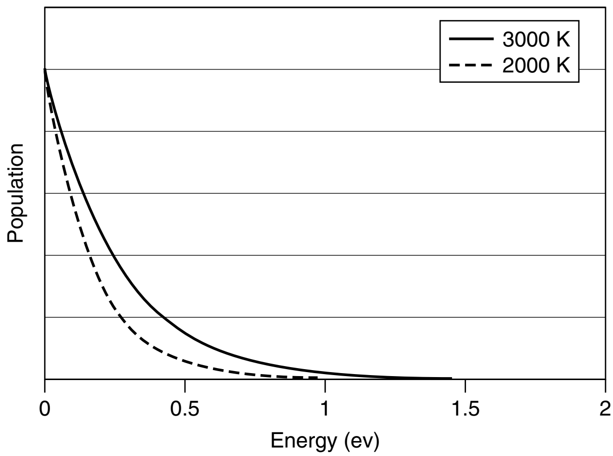

Figure 1.9.1 Generalized Boltzmann distribution of atomic energies.

1.602×10^{-19} J of energy). The number of atoms at this higher-energy state is on the order of 10^{-30}! It is fair to say that there are essentially no atoms with energy this high at room temperature (and this energy level isn't really very high at all; ions have much higher energies). Now, if the temperature is increased to, say, 5000 K, the population of atoms at this energy level increases to 2×10^{-2} or 2%, a sizable concentration at this level which will lead to the emission of light when these atoms lose their energy and return to ground state. The experience of observing red-hot and white-hot objects such as a stove element or glowing metal shows us that anything heated to beyond 1000 K will emit light in the form of a glow.

In a system such as we have described, it is assumed that there are no external sources of energy, so the population of atoms at any given energy level is governed solely by temperature. It is then said to be in *thermal equilibrium*. Examining the formula, we can see that if the temperature of the entire system were raised, the distribution would shift and more atoms would reach higher energies; however, the population of a lower level will always exceed that of a higher level. Higher temperatures result in increased populations of atoms at higher energy levels, and this results in larger energies. Light-producing transitions in such a substance will also have more energy, so more emitted photons from hotter objects will have more energy (and hence be shifted toward the blue region of the spectrum) than the photons emitted by cooler objects. It must also be noted that only at 0 K (absolute zero) will all atoms be at ground state (the lowest energy level).

1.10 QUANTUM VIEW OF BLACKBODY RADIATION

A blackbody absorbs all radiation incident upon it, and emission from such an object depends solely on the temperature of the body. At hotter temperatures it tends to

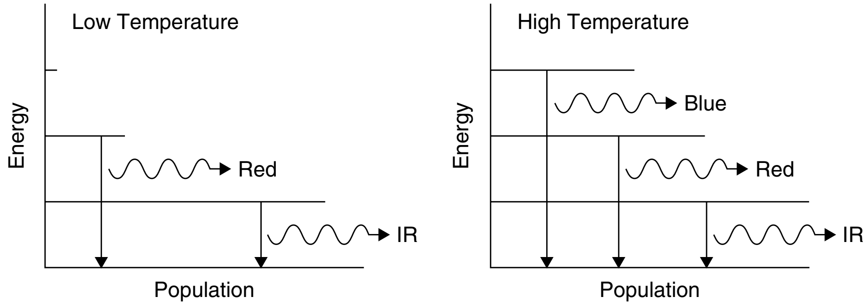

Figure 1.10.1 Energy-level populations at various temperatures.

have a higher blue content, and at lower temperatures the emission is seen as red. Examining the energy levels in this situation (as in Figure 1.10.1), we see that at low temperatures only low energy levels will be excited. The atoms in this case will have enough energy to cause transitions creating a reasonable amount of infrared light and some red light, but very few atoms will have enough energy to allow the production of blue light. As the temperature of the object is increased, higher atomic levels will be excited and blue emissions will be seen. This shift is due to the fact that blue light is literally more energetic than red light, as evident from the Planck relationship $E = h\nu$.

It may also occur to the reader that blackbody radiation is *broadband*. Unlike line spectra produced by most gas discharges, thermal light tends to occur in the form of a continuum of wavelengths spanning a large range. The answer is in the spacing of energy levels in the emitting substance. Gases usually have well-defined discrete wavelengths, due to the fact that each atom in the gas is ultimately identical and completely independent (i.e., no interactions between the atoms in the gas lead to changes in the discrete atomic energy levels). In solids, atoms are closely packed and interact with each other, which serves either to widen energy levels to the point where they overlap or create new electron energy levels. This leads us to speak instead of the energy levels in solids as *energy bands*. In some cases these bands are simply a discrete level that has been broadened (i.e., spans a range of possible energies) through various mechanisms. If the discrete energy levels in Figure 1.10.1 were replaced by wide bands of possible energies, it is easy to see that the average light emitted will be broadband.

1.11 BLACKBODIES AT VARIOUS TEMPERATURES

To get an idea of how blackbody radiation depends on temperature, consider blackbody radiation curves for an object at various temperatures, as shown in Figure 1.11.1. Consider first an extremely hot object at 6000 K. Although a great deal of radiation is emitted in the infrared region of the spectrum, the vast majority

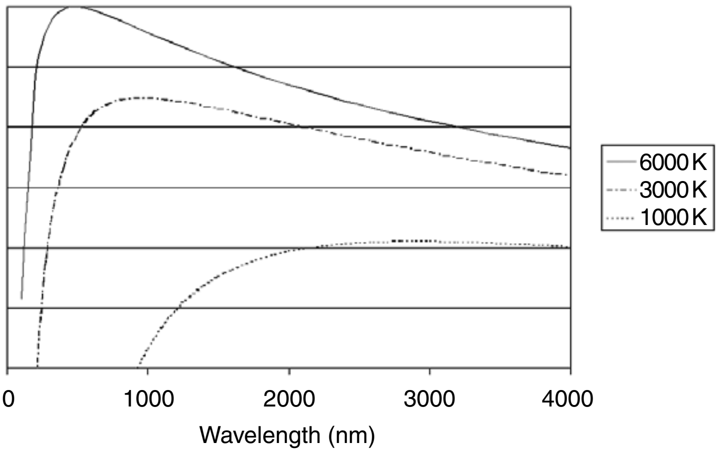

Figure 1.11.1 Blackbody radiation spectra.

lies in the visible range and more specifically, in the blue region of the visible spectrum. Such an object is said to have a *color temperature* of 6000 K and approximates that produced by sunlight. Many commercial lamps, especially those used by photographers, have a color temperature rating like this; it is simply a comparison to an equivalent output from a blackbody radiator.

Considering a somewhat cooler object at 3000 K, the object appears red when viewed since there is a great deal more red light emitted than blue light. Incandescent lamps often have a color temperature around 3500 K, and photographs taken under such light appear reddish yellow and require a blue correcting filter to have accurate color reproduction. This point is also illustrated in the differences between light from a regular incandescent lamp and a halogen lamp. In halogen lamps the filament burns in an atmosphere of halogen gas (e.g., iodine) which allows the filament to burn much hotter than in an ordinary incandescent light bulb without burning out prematurely. The increased temperature results in a shift from the reddish-yellow (warmer) light of an ordinary incandescent lamp to a much whiter light (i.e., containing more blue). Halogen lighting is therefore often used for illumination of objects where truer color rendition is required (e.g., display of artwork). As this object cools to, say, 1000 K, the emission of light shifts even further into the infrared, and only a small amount of red light is emitted (with almost no blue). Such an object is said to be "red hot" and the vast majority of its emission is in the infrared region between 2- and 6-μm wavelengths. The red we see is only a tiny fraction of the total radiation emitted.

Objects at room temperature (300 K) also emit radiation in the infrared centered at about 10 μm. In fact, even objects at a cold 3 K (the background temperature of the universe found in deep outer space) emit radiation at 3-mm wavelengths. Such long wavelengths are in the microwave region of the spectrum and can be detected using sensitive microwave receivers. Cosmologists often call this microwave background the “leftovers” from the “big bang” that created the universe.

1.12 APPLICATIONS

The fact that every object emits radiation can be exploited for a number of uses. For example, the human body at a temperature of about 312 K emits a large amount of infrared radiation centered about a wavelength of 10 μm. This is used for security and convenience lighting purposes by passive infrared (PIR) motion detectors such as that in Figure 1.12.1. Lenses focus radiation from the area under surveillance onto a sensitive pyroelectric detector that detects changes in the incident radiation. Pyroelectric detectors are made from ferroelectric crystals and work by allowing incident radiation to change the temperature of the crystal, which affects the charge on the electrodes. They have a wide response range of 1 to 100 μm and are quite rapid at detecting change. As a human (or any other warm object) walks across the area in front of the detector, the amount of radiation received changes suddenly, triggering a light or alarm.

It may be worth mentioning that clothing and other items normally opaque to visible light are often quite transparent to far-infrared light. To this end a commercially viable security camera is available which, quite literally, has the ability to see through clothing. In this age of tight airport security, this would make it quite impossible to conceal a weapon of any type, including those that do not affect metal detectors. Of course, the privacy ramifications have not yet been dealt with fully. This technology has already been used by border patrols to identify illegal aliens hiding in concealed compartments of trucks. The IR radiation emitted from the humans inside is easily detected by an IR camera, which clearly

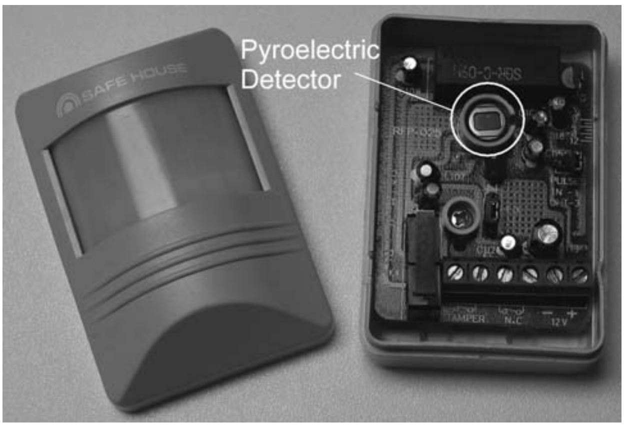

Figure 1.12.1 Passive infrared detector.

shows the outlines of the (warm) figures concealed inside. More mundane applications of blackbody radiation include noncontact thermometers used worldwide by the medical community. These in-the-ear thermometers measure the IR radiation emitted from the eardrum (which is inside the body and hence accurately reflects the body's core temperature) and calculate body temperature from this.

1.13 ABSORPTION AND COLOR

We have seen that atoms at low-energy states can absorb energy and that this energy can be light. If an atom at a low-energy state absorbs a photon with a particular amount of energy, that atom will reach a high-energy state with a final energy equal to its original energy plus the energy of the photon absorbed. This is why substances appear in certain colors. For example, a red liquid appears red when you look through it because red photons are not absorbed by the liquid as are other colors of light (such as blue, green, and yellow). The same is true of a blue liquid, in which blue photons are not absorbed but greens, yellows, and reds are. So why does the blue liquid absorb the lower-energy red photon yet allow the higher-energy blue photon through? The answer lies in the nature of absorption itself, in which the energy of an absorbed photon must match the energy of a transition: If energy levels in a particular atom exist 2.1 eV apart, a 2.1-eV photon is required to excite the atom from the lower level to the upper level. A photon with 1.7 eV of energy falls short of the energy needed to make the jump, whereas a 2.3-eV photon has too much energy.

Another way of looking at absorption is as the opposite of emission. This is expected in nature: If atoms can emit light to lose energy, they must be able to absorb it to gain energy. Because of the quantized nature of energy levels, it is possible to find atomic and molecular species that have energy levels allowing the absorption of photons of essentially any energy. In the case of an atom such as hydrogen, energy levels are specific and sharply defined; however, molecules have broad energy bands that allow absorption (or emission) over a wide spectrum of wavelengths. This is why liquids absorb a range of wavelengths (such as all red and orange light) instead of a specific wavelength such as a low-pressure gas would. This entire concept of atomic emission is dealt with in detail in Chapter 2.

1.14 EFFICIENCY OF LIGHT SOURCES

Since the majority of light sources (natural or artificial) are blackbody radiators, this discussion would not be complete without a discussion of efficiency of various sources. Most practical sources of light involve a heated medium, and as such, most of the radiation emitted is in the infrared region of the spectrum at wavelengths too long to be usable by the eye as light. Some sources are more efficient than others at emitting light at visible wavelengths. To compare light output by various sources, we use a unit called a *lumen*, which represents the power of light emitted by a source.

In terms of a practical light source, we measure efficiency as a function of light output to power input using the unit *lumens per watt*, which measures the number of lumens a light source emits for 1 W of input power. In theory, a perfect light source emitting at the peak sensitivity of the human eye could deliver 622 lm/W. Practical light sources fall far below this figure.

Consider an oil lamp in which kerosene is drawn up through a wick and burned. This is an incandescent system in which heat from the flame causes particles of carbon to burn brightly in the flame. Although kerosene contains a large amount of energy in a given volume, it burns inefficiently, with an efficiency of about 0.1 lm/W. The vast majority of energy in the kerosene is converted to heat rather than light. Electric incandescent lamps with a tungsten filament fare better, with an efficiency of up to 20 lm/W.[3] This still represents an enormous waste of energy, though, since over 90% of the emission from such a lamp is wasted as heat. Other artificial lamp technologies such as fluorescent lamps (discussed in Chapter 2), provide higher efficiencies of about 90 lm/W, and discharge lamps such as the sodium lamp (often used for street lights and easily identified by the yellow color of the light emitted) have the highest efficiencies, at about 150 lm/W.[3] Bear in mind that these last two technologies (fluorescent and sodium lamps) are not thermal light sources at all but atomic emission sources in which electrical energy is converted directly to light, not heat, as in the incandescent lamp.

PROBLEMS

1.1 Halogen lamps burn a tungsten filament in an atmosphere consisting of a halogen gas such as iodine or bromine. Explain why these lamps are (**a**) more efficient and (**b**) produce a whiter light than that of regular incandescent tungsten lamps. Specifically, obtain temperature figures (manufacturers' Web sites often list this) of these lamps and calculate the output wavelength and power for an incandescent and a halogen lamp.

1.2 Cool sources have a distinct red shift, whereas hotter sources appear yellow. Is there a temperature for an emission source at which all visible colors are equally balanced? If not, what is the optimal temperature at which output in the red and output in the blue are at the same level. At this optimal temperature, how much brighter would the yellow output be?

1.3 Passive infrared detectors (PIRs) use the blackbody emission from people to detect their presence. One problem with the use of such detectors in a security system is that pets such as cats can trigger these detectors. Knowing that a cat

[3] The values quoted for efficiencies of light sources are recent (2000) and represent midpoints for a given class of lamp. Efficiency of artificial light sources has improved drastically since Edison's first electric incandescent lamp of 1880, which boasted an efficiency of 1.6 lm/W! Today, the average modern incandescent lamp yields an efficiency of about 10 times that value. Other lamp technologies, including the fluorescent lamp, discovered in 1935, have also improved efficiency over the years.